

$$1. \quad T(n) = 3T(n-1) - 15$$

$$T(1) = 8$$

$$- \quad 3(3T(n-2) - 15) - 15 \quad T(n-1)$$

$$- \quad 3(3(3T(n-3) - 15) - 15) - 15 \quad T(n-2) \quad 3^3 T(n-3) - (n-1)15$$

$$T(1) + 3 \sum_{i=2}^{n-1} 15(n-1) \quad (-4)$$

$$2. \quad T(n) = T(n-1) + n - 1$$

$$T(1) = 3$$

$$- \quad T(n-1) \quad (T(n-2) + n-1) + n-1$$

$$- \quad T(n-2) \quad ((T(n-3) + n-1) + n-1) + n-1$$

$$T(1) + \sum_{i=2}^{n-1} n-1 \quad (-3)$$

$$3. \quad T(n) = 6T(n/6) + 2n + 3$$

$$T(n) = 1$$

$$- \quad T(n-1) \quad 6(6T(n/6) + 2n + 3) + 2n + 3, \quad n \text{ is a power of } 6$$

$$6^{n-1} \sum_{i=2}^{n/6} 2n + 3 \quad (-4)$$

Brandon Buchanan

4. $T(n) = 4T(n/3) + 2n - 1$ for n a power of 3
 $T(1) = 2$

$$T(1) + 4^{n-1} \sum_{i=2}^{n/3} 2n-1$$

5. $a = 5, b = 4, f(n) = n - 3$

case 1 $n - 3 = O(n^{\log_4 5 - \epsilon}), \epsilon > 0$, false

case 2 $n - 3 = \Theta(n^{\log_4 5})$, false

case 3 $n - 3 = \Omega(n^{\log_4 5 + \epsilon}), \epsilon > 0$

-3

6. $a = 9, b = 3, f(n) = n^2 + 4$

case 1 $n^2 + 4 = O(n^{\log_3 9 - \epsilon}), \epsilon > 0$

$n^2 + 4 = O(n^{2 - \epsilon}), \epsilon > 0$

case 2 $n^2 + 4 = \Theta(n^2)$, true

3 false

7. $a = 6, b = 2, f(n) = n^2 - 2$

case 1 $n^2 - 2 = O(n^{\log_2 6 - \epsilon}), \epsilon > 0$, false

case 2 $n^2 - 2 = \Theta(n^{\log_2 6}), \epsilon > 0$, false

case 3 $n^2 - 2 = \Omega(n^{\log_2 6 + \epsilon}), \epsilon > 0$

-3

8. $a=4, b=2, f(n^3+7)$

case 1 $n^3+7 = O(n^{\log_2 4 - \epsilon})$, $\epsilon > 0$, false

case 2 $n^3+7 = O(n^{2-\epsilon})$, $\epsilon > 0$, false

case 3 $n^3+7 = \Omega(n^{2+\epsilon})$, $\epsilon > 0$

$4\left(\frac{n^3+7}{2}\right) = 2n^3+14 \leq n^3+7$, true

10. $T(n^x) = n \cdot T(n^{x-1})$ power
 $T(n^{x-1}) = n \cdot n \cdot T(n^{x-2})$

11. $T(n^y) = T(n^{y/2}) \times T(n^{y/2})$, y is even
 $T(n^y) = T(n^{y/2}) \cdot n \cdot T(n^{y/2})$, y is odd